

Introduction to Electronic Design Automation

Homework 2

(Problems 6-8)

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6 Two-level Logic Minimization

6. (a)

Use variable order $abcde$.

$$G = abce + abde + ab\bar{c}e + \bar{a}b\bar{c}e + \bar{a}\bar{b}\bar{c}\bar{e} + \bar{a}\bar{c}\bar{d}e + bc\bar{d}e + cd\bar{e}$$

$$D = \bar{a}b\bar{d}e + abcd\bar{e} + ac\bar{e}$$

Expanding them into minterms:

$$G = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 25, 27, 28, 29, 31\}$$

$$D = \{9, 13, 20, 22, 28, 29, 30\}$$

Thus, we have the minterms of onset F as:

$$F = G - D = \{0, 1, 2, 4, 11, 12, 25, 27, 31\}$$

Therefore,

$$F \cup D = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 22, 25, 27, 28, 29, 30, 31\}$$

$$\begin{aligned} R &= \{0, 1, \dots, 31\} - (F \cup D) \\ &= \{3, 5, 6, 7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24, 26\} \end{aligned}$$

Writing them as product terms:

$$\begin{aligned} R &= \bar{a}\bar{b}\bar{c}de + \bar{a}\bar{b}c\bar{d}e + \bar{a}\bar{b}cd\bar{e} + \bar{a}\bar{b}cde + \bar{a}b\bar{c}\bar{d}\bar{e} + \bar{a}b\bar{c}d\bar{e} \\ &\quad + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}\bar{d}\bar{e} + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} \\ &\quad + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} \end{aligned}$$

6. (b)

First, we apply the Quine–McCluskey tabular method on $F \cup D$ to generate all prime implicants.

Round 1	Round 2	Round 3
$\bar{a}\bar{b}\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{b}\bar{c}\bar{d}$ $\bar{a}\bar{b}\bar{c}\bar{e}$ $\bar{a}\bar{b}\bar{d}\bar{e}$	
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}b\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{c}\bar{d}\bar{e}$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}c\bar{d}\bar{e} \checkmark$	
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}b\bar{c}\bar{d}\bar{e} \checkmark$ $a\bar{b}\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{b}\bar{c}e \checkmark$ $\bar{a}bde \checkmark$ $\bar{b}\bar{c}\bar{d}e \checkmark$ $\bar{a}bcd \checkmark$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}b\bar{c}e \checkmark$ $ac\bar{d}\bar{e} \checkmark$	$c\bar{d}\bar{e}$ $b\bar{d}e$ $b\bar{c}e$ bcd $ac\bar{e}$
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}b\bar{c}d\bar{e} \checkmark$ $a\bar{b}\bar{c}d\bar{e} \checkmark$ $a\bar{b}c\bar{d}\bar{e} \checkmark$ $ab\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $ac\bar{e} \checkmark$ $ab\bar{c}e \checkmark$ $ab\bar{d}e \checkmark$ $abcd \checkmark$ $abc\bar{e} \checkmark$	abe abc
$ab\bar{c}d\bar{e} \checkmark$ $abc\bar{d}\bar{e} \checkmark$ $abcd\bar{e} \checkmark$	$abde \checkmark$ $abce \checkmark$ $abcd \checkmark$	
$abcde \checkmark$		

Table 1: Quine–McCluskey tabular method for generating prime implicants

Thus, the prime implicants are

$$\{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{b}\bar{c}\bar{e}, \bar{a}\bar{b}\bar{d}\bar{e}, \bar{a}\bar{c}\bar{d}\bar{e}, c\bar{d}\bar{e}, b\bar{d}e, \bar{b}\bar{c}e, bcd, ac\bar{e}, abe, abc\}.$$

Next, we use the Column covering of Boolean matrix to find the essential prime implicants. The Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{c}\bar{e}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}\bar{e}$	$c\bar{d}\bar{e}$	$b\bar{d}e$	$\bar{b}\bar{c}e$	bcd	$ac\bar{e}$	abe	abc
$m_0 = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$	1	1	1	0	0	0	0	0	0	0	0
$m_1 = \bar{a}\bar{b}\bar{c}d\bar{e}$	1	0	0	1	0	0	0	0	0	0	0
$m_2 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	1	0	0	0	0	0	0	0	0	0
$m_4 = \bar{a}\bar{b}cd\bar{e}$	0	0	1	0	1	0	0	0	0	0	0
$m_{11} = \bar{a}b\bar{c}d\bar{e}$	0	0	0	0	0	0	1	0	0	0	0
$m_{12} = \bar{a}bcd\bar{e}$	0	0	0	0	1	0	0	1	0	0	0
$m_{25} = ab\bar{c}\bar{d}\bar{e}$	0	0	0	0	0	1	1	0	0	1	0
$m_{27} = ab\bar{c}d\bar{e}$	0	0	0	0	0	0	1	0	0	1	0
$m_{31} = abcde$	0	0	0	0	0	0	0	0	0	1	1

Table 2: Boolean matrix

6. (c)

Rows m_2 and m_{11} each have only one 1, so $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$ are essential prime implicants. So we remove the rows m_2, m_{11} and the columns of $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$. We also remove the rows that have 1 in the removed columns, which are m_0, m_{25}, m_{27} . After removing these rows, the columns $a\bar{c}\bar{e}$ and $b\bar{d}e$ contain only zeros, so we remove them as well. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	bcd	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	0	1	0	0	0	0
$m_4 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	1	0	1	0	0	0
$m_{12} = \bar{a}bc\bar{d}\bar{e}$	0	0	0	1	1	0	0
$m_{31} = abcde$	0	0	0	0	0	1	1

Table 3: Reduced Boolean matrix

6. (d)

Column $c\bar{d}\bar{e}$ covers rows m_4 and m_{12} . It dominates both columns $\bar{a}\bar{b}\bar{d}\bar{e}$ and bcd , so we remove these two dominated columns. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0	0
$m_4 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	0	1	0	0
$m_{12} = \bar{a}bc\bar{d}\bar{e}$	0	0	1	0	0
$m_{31} = abcde$	0	0	0	1	1

Table 4: Reduced Boolean matrix

Since rows m_4 and m_{12} have only one 1, the prime implicant $c\bar{d}\bar{e}$ is an induced essential prime implicant. Therefore, we select it and remove rows m_4 and m_{12} .

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0
$m_{31} = abcde$	0	0	1	1

Table 5: Reduced Boolean matrix

6. (e)

After the reductions in (d), there is no cyclic core to solve, since the remaining matrix is decomposed into two independent single-row choices:

$$m_1 : \{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{c}\bar{d}e\}$$

$$m_{31} : \{abe, abc\}$$

Thus, the independent set heuristic is not applicable. We only need to choose one prime implicant from each set. For example, we choose

$$\bar{a}\bar{b}\bar{c}\bar{d} \quad \text{and} \quad abe.$$

Therefore, the selected prime implicants are

$$\bar{a}\bar{b}\bar{c}\bar{e}, \quad b\bar{c}e, \quad c\bar{d}\bar{e}, \quad \bar{a}\bar{b}\bar{c}\bar{d}, \quad abe.$$

6. (f)

There are four minimum covers with 5 prime implicants:

$$G_1^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abe.$$

$$G_2^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abe.$$

$$G_3^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abc.$$

$$G_4^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abc.$$

7 Static Timing Analysis**7. (a)**

In this part, we construct the circuit as a DAG, where each node $v \in V$ represents a gate and each edge $e \in E \subseteq V \times V$ represents a connection between gates. Let $R(v)$ denote the required time at the input of node v , $A(v)$ denote the arrival time at the output of node v , and $d_{u,v}$ denotes the delay from node u 's output to node v 's output. Also, we define the slack of node v as

$$S(v) = R(v) - A(v).$$

Suppose there is a node v such that

$$S(v) = R(v) - A(v) \leq c$$

We want to show that there exists a path satisfying every node on the path has slack at most c from some primary input to v and from v to some primary output.

1. From some primary input to v

First, if v is a primary input, then we are done. Otherwise, since

$$A(v) = \max_{u:(u,v) \in E} (A(u) + d_{u,v})$$

there exist a node u such that there is an edge from u to v and

$$A(v) = A(u) + d_{u,v}.$$

And since

$$R(u) = \min_{w:(u,w) \in E} (R(w) - d_{u,w}) \leq R(v) - d_{u,v},$$

we have

$$S(u) = R(u) - A(u) \leq R(v) - d_{u,v} - A(u) = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on u and repeat this process until we reach a primary input.

2. From v to some primary output

First, if v is a primary output, then we are done. Otherwise, since

$$R(v) = \min_{w:(v,w) \in E} (R(w) - d_{v,w})$$

there exist a node w such that there is an edge from v to w and

$$R(v) = R(w) - d_{v,w}.$$

And since

$$A(w) = \max_{x:(x,w) \in E} (A(x) + d_{x,w}) \geq A(v) + d_{v,w},$$

we have

$$S(w) = R(w) - A(w) \leq R(v) + d_{v,w} - A(v) - d_{v,w} = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on w and repeat this process until we reach a primary output.

Hence, there exists a path from some primary input to v and from v to some primary output such that the slack of each node on these paths is at most c .

8 Technology Mapping